

Anticipating Second-Order Dynamics in Protocol Design through Adversarial LLM Simulation

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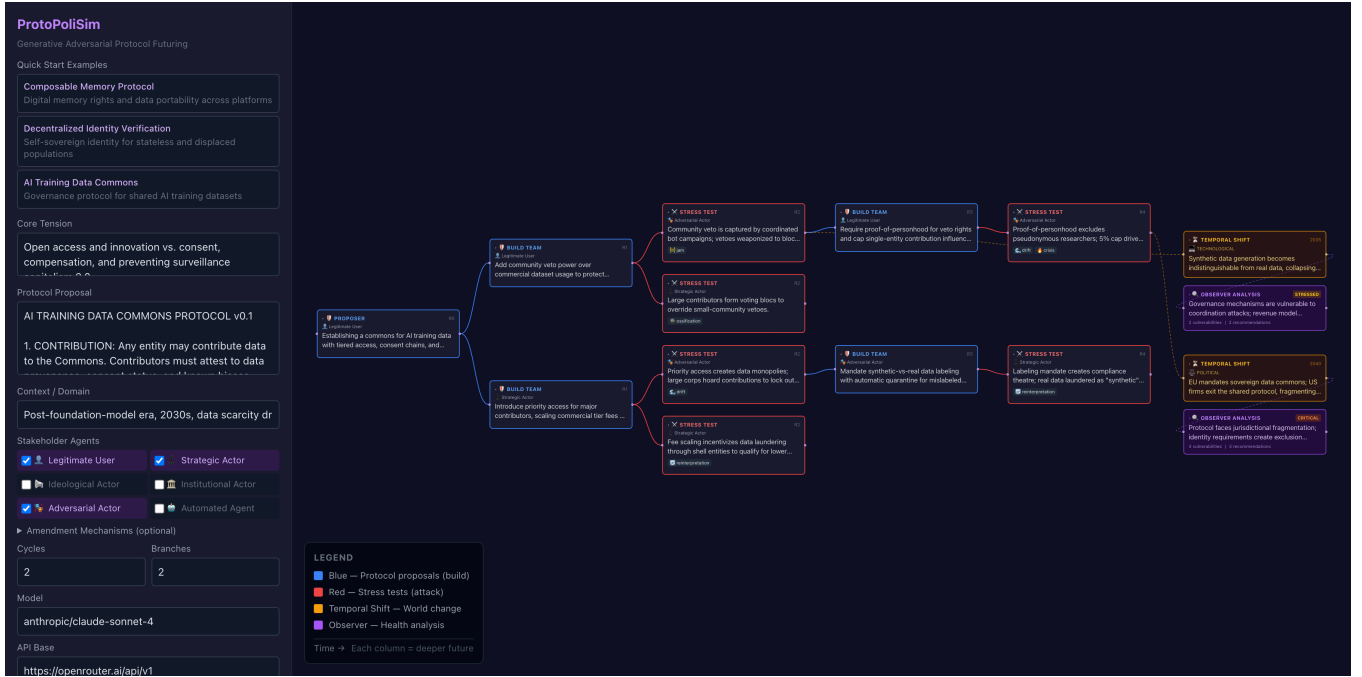


Figure 1: The ProtoPoliSim prototype during a two-cycle adversarial simulation of an AI training data commons protocol. The sidebar (left) provides configuration for protocol proposals, core tensions, and stakeholder persona types. The canvas (right) renders the branching future cone as an interactive tree: blue nodes are build-team proposals, red nodes are adversarial stress-tests annotated with detected second-order dynamics (drift, jam, ossification), amber nodes introduce temporal shifts, and purple nodes present observer health analyses. Time flows left to right; each column represents a deeper speculative future.

Abstract

Governance protocols—the rules, norms, and technical standards that coordinate collective action—are notoriously difficult to evaluate before deployment. Once enacted, they may *drift* from intent, become *jammed* by adversarial actors, or *ossify* against necessary change. These second-order dynamics emerge from the situated reasoning of actors who were not in the room when the rules were written, making them invisible to conventional testing. We argue that large language models, grounded in deep synthetic personas, can serve as a pre-deployment stress-testing mechanism for governance protocols—surfacing identity-specific failure modes that neither classical agent-based modeling nor participatory design workshops can reach alone. We operationalize this argument through **ProtoPoliSim**, a working prototype that instantiates persona-driven

agents inside the *Protocol Futuring* adversarial cycle—a structured Blue (build) / Red (attack) / Temporal Shift / Observer pipeline—to automatically elicit and classify second-order dynamics as structured simulation outputs. Each run produces a branching tree of protocol futures, with every node annotated for drift, jam, ossification, or reinterpretation. We present the system’s architecture, demonstrate its interface, and discuss the open methodological and validation challenges this approach poses for the PoliSim community.

CCS Concepts

• **Human-centered computing** → Collaborative and social computing design and evaluation methods.

Keywords

Protocol Design; Synthetic Personas; Social Simulation; LLM Agents; Second-Order Dynamics; Design Futuring; Governance

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1 Introduction

A growing body of work in policy HCI [24, 25] and computational governance [14] is shifting attention from policy-making—the crafting of rules through deliberation and institutional authority—toward *protocol design*: the engineering of sociotechnical infrastructures that embed values, power relations, and behavioral assumptions directly into technical media [5, 13]. Protocols are not policies. They are “engineered arguments” [22]: an HTTP header enforces a privacy stance; a token-weighted voting mechanism encodes a theory of legitimacy; a data-commons license operationalizes a position on access and compensation. As infrastructure studies has long recognized, these embedded decisions “sink into” [21] the background of practice, becoming invisible precisely as they become consequential [20]. The stakes are correspondingly high: “*the fate of our world may rest on protocols*” [3], yet the failure modes that matter most are the ones the designer never modeled.

Yet the central difficulty of protocol design is that the most consequential failures are not first-order but *second-order*: they emerge not from the protocol’s immediate function but from the ways actors adapt to, game, and reinterpret the rules once deployed [12]. A data-commons charter is exploited by actors who were not envisioned at design time. A moderation policy drifts as communities reinterpret its norms. An AI governance framework ossifies around the incumbents it was meant to regulate. These dynamics—*drift*, *jam*, and *ossification*—are what Frederik Pohl’s maxim captures: a good forecast should predict “*not the automobile but the traffic jam*” [19]. Goodhart’s Law describes one mechanism: “*when a measure becomes a target, it ceases to be a good measure*” [10, 15]—but the gaming that triggers this effect is performed by specific kinds of actors with specific interests, cultural contexts, and institutional positions. Once institutionalized, protocols resist change even when their founding rationales no longer apply [1]. The designer’s problem is therefore not merely functional correctness but *anticipatory robustness*: can the protocol withstand the adaptive behavior of the diverse actors it will encounter?

Existing approaches to this problem face a trilemma. Classical agent-based modeling (ABM) [4, 9] and Monte Carlo simulation frameworks such as cadCAD [27] provide systematic repeatability and scale, but require detailed mathematical models of the very behaviors the designer is trying to anticipate—and second-order dynamics are, by definition, what the designer *did not* model. These approaches can stress-test known parameters, but cannot generate the qualitatively novel responses, cultural reinterpretation, or situated reasoning that drives real institutional gaming. Design fiction and participatory workshops [7, 24] generate genuine human depth and surface culturally specific dynamics, but require organizing sessions with domain experts whose availability, disciplinary perspectives, and institutional positions inevitably shape—and limit—the failure modes that get surfaced; the results are only as diverse as the experts in the room, and lack systematic adversarial pressure that could push beyond the participants’ shared assumptions. Generic LLM agent frameworks improve fluency but typically instantiate agents as thin stakeholder labels (“*Corporate Actor*”, “*Frustrated User*”)—abstractions too shallow to produce the identity-grounded exploits that protocol stress-testing demands.

Our position is that this trilemma is resolvable. We argue that **large language models, grounded in deep synthetic personas and structured by an adversarial simulation cycle, can serve as a pre-deployment stress-testing mechanism for governance protocols**—surfacing second-order dynamics that the protocol designer did not intend and could not have anticipated from first principles alone. The key enabling insight is that recent advances in synthetic persona generation [16, 23] make it possible to instantiate LLM agents with genuine human depth—hundreds of attributes spanning demographics, values, institutional affiliations, and behavioral dispositions—while the Protocol Futuring methodology [12] provides the adversarial cycle structure (Blue build / Red attack / Temporal Shift / Observer) that converts those personas from passive role-players into active protocol stress-testers. Neither element suffices alone: deep personas without adversarial structure produce rich but undirected narratives; adversarial structure without persona depth produces stereotyped attack patterns. Together, they constitute a simulation paradigm that is simultaneously richer than classical ABM, more systematic than design fiction, and more scalable than participatory workshops.

To test whether LLM-driven simulation can meaningfully perform Protocol Futuring—that is, whether persona-grounded agents can generate second-order dynamics that are non-obvious, structurally diverse, and useful to protocol designers—we built **ProtoPoliSim**, a working, open-source prototype (Figure 1) presented alongside a companion full paper at CHI 2026 [12]. The prototype is deliberately a research instrument, not a finished tool: its purpose is to make the open question of LLM-driven protocol stress-testing concrete and inspectable, so that the community can evaluate where this approach succeeds, where it falls short, and what validation frameworks it requires. In the remainder of this paper, we situate our approach in the converging literatures on LLM-driven social simulation and policy HCI (§2), describe the prototype’s design as a research through design inquiry (§3), and discuss limitations and open challenges (§4).

2 Background

2.1 Social Simulation for Policy: From ABM to LLM Agents

Agent-based modeling has long served as the primary computational approach to policy simulation, from Epstein and Axtell’s foundational work on emergent social phenomena [9] to Bonabeau’s general methodology for complex human systems [4] and Epstein’s philosophy of “generative social science” [8]. However, classical ABM faces a fundamental limitation: agents defined by fixed behavioral rules cannot generate qualitatively novel responses or culturally situated reasoning.

The emergence of large language models has opened a new frontier. Park et al. [17] demonstrated that LLM-powered generative agents can produce human-like social behavior without explicit behavioral rules, and in subsequent work showed that generative agent simulations of 1,000 real individuals can replicate their attitudes and behaviors with remarkable fidelity—reducing accuracy biases across racial and ideological groups compared to agents given only demographic descriptions [18]. Argyle et al. [2] showed that conditioned LLMs can simulate diverse human subpopulations,

reproducing survey response distributions—a capacity they term “silicon sampling.” Yu et al. [26] extended this to political decision-making at population scale. These advances have catalyzed a growing community around LLM-driven political simulation, exemplified by the PoliSim workshop at CHI 2026 [14].

2.2 Protocol Futuring and Policy HCI

The second strand concerns HCI’s growing engagement with policy as designable infrastructure. Yang et al. [25] articulate a vision for HCI-policy collaboration in which design methods—prototyping, participatory inquiry, iterative testing—are applied to the policy domain itself, treating policies as designable and testable artifacts rather than fixed legislative outputs. This framing resonates with the broader turn in HCI toward *infrastructural speculations* [24]: speculative design practices that move beyond individual artifacts to interrogate the protocols, standards, and governance structures that constitute the built environment of collective life.

Protocol Futuring [12] is a methodological framework that operationalizes this turn by centering governance protocols—rather than artifacts or actors—as the primary material of speculative inquiry. Drawing on infrastructure studies [20, 21], it organizes inquiry around iterative handoffs between design (blue) and disruption (red) phases, forcing protocols to encounter adversarial pressure across simulated temporal horizons [6]. The adversarial relay structure is specifically designed to surface second-order dynamics—drift, jam, ossification, crisis, reinterpretation—that infrastructure studies has identified as characteristic of long-lived sociotechnical systems [1]. Prior Protocol Futuring exercises have been conducted as facilitated workshops with human participants [11], generating rich but slow and difficult-to-scale outcomes. ProtoPoliSim automates this cycle with LLM agents, preserving the method’s adversarial structure while gaining the repeatability, speed, and systematic variation that computational simulation affords.

3 ProtoPoliSim: Research through Design

Following the research through design tradition in HCI [28], we built ProtoPoliSim as a working, open-source prototype¹ whose primary purpose is not to deliver a finished product but to investigate a research question: *how well can LLM agents, structured by an adversarial cycle and grounded in synthetic personas, surface second-order dynamics that protocol designers did not anticipate?* The prototype is the instrument through which we interrogate this question; its design choices embody our theoretical commitments and make them available for critique. Built with Next.js, React Flow, and the OpenRouter API, the system provides an interactive environment for configuring and running adversarial protocol simulations (Figure 1). Each simulation begins with a user-specified *protocol proposal* and *core tension* (e.g., an AI training data commons; tension: open access vs. consent and compensation), and proceeds through repeating four-phase cycles:

Blue Phase (Build). Blue-team personas—instantiated as deep synthetic profiles anchored in specific stakeholder positions (e.g., *an open-source contributor in a low-resource country who depends on the commons for training data access*)—propose protocol refinements that serve their identity-grounded interests. Each persona’s

response is conditioned on its full attribute profile, the accumulated world-state, and the history of prior protocol moves.

Red Phase (Attack). Red-team personas—whose profiles encode adversarial, strategic, or institutionally resistant positions—stress-test each blue proposal. Each red agent produces: (i) an updated protocol representing the gamed or distorted version; (ii) a proposal narrative explaining the exploit; and (iii) a set of *typed second-order dynamics* drawn from the Protocol Futuring ontology: drift, jam, ossification, crisis, or reinterpretation. This structured output is the elicitation mechanism through which the system surfaces dynamics that were not designed into the protocol.

Temporal Shift. An exogenous perturbation agent introduces world-state shocks (technological, political, ecological) that shift the environment for the next cycle, modeling how protocols encounter futures that no design-time persona could have anticipated.

Observer Analysis. A fourth-position agent reads the full simulation history and produces a structured report: identified vulnerabilities, cross-cycle pattern analysis, design recommendations, and a protocol health classification across a four-level taxonomy (healthy / stressed / critical / failing).

Persona-Grounded Agents. ProtoPoliSim integrates recent advances in synthetic persona generation [16, 23] to ground every blue and red agent in a typed stakeholder persona. The system ships with a default taxonomy of six actor types—legitimate users, strategic actors, ideological actors, institutional actors, adversarial actors, and automated agents—but the taxonomy is user-configurable: designers can add, remove, or redefine stakeholder roles to match the protocol under test. Rather than prompting an LLM with a thin label (“Corporate Actor”), the system constructs each agent’s prompt from a structured profile encoding a specific subject position, institutional logic, and adversarial disposition. This draws on Paglieri et al.’s insight that persona diversity should optimize for *support coverage* (what is possible) rather than *density matching* (what is most probable) [16]—directly relevant to stress-testing, where the most dangerous failure modes are triggered by actors whose trait combinations are rare but not impossible.

Branching Future Cone. Each blue and red node branches from all frontier nodes of the prior cycle, producing an expanding tree of possible protocol futures visualized as an interactive left-to-right future cone. The branching structure makes visible not just what could happen, but the full space of adversarial trajectories the protocol might traverse—a design space of anticipation rather than a single predicted outcome.

4 Discussion

Contribution and relation to prior work. The fundamental limitation of existing computational approaches to protocol stress-testing—whether ABM [4, 9] or Monte Carlo frameworks like cadCAD [27]—is that they require the designer to specify mathematical models of the behaviors being tested. But second-order dynamics are precisely what the designer *cannot* model in advance: you cannot write a differential equation for an actor who reinterprets your protocol’s norms in a way you never imagined. ProtoPoliSim’s contribution is to replace formal behavioral specifications with LLM agents whose responses emerge from situated

¹Source code: <https://github.com/realitydeslab/protopolisim>

reasoning over deep persona profiles, structured by an adversarial cycle that *forces* the generation of protocol failures. This produces structured, comparable outputs—every node annotated with typed second-order dynamics, every cycle yielding a protocol health report—while operating at a level of qualitative richness that mathematical simulation cannot reach. Compared to Park et al.’s generative agents [17, 18], which simulate open-ended social behavior, ProtoPoliSim constrains agents within an adversarial structure specifically designed to surface protocol failure modes. Compared to Argyle et al.’s “silicon sampling” [2] and Yu et al.’s political simulations [26], which reproduce aggregate population-level distributions, our approach prioritizes adversarial depth over demographic breadth—seeking the rare actor who breaks the protocol rather than the median citizen who follows it. And compared to workshop-based Protocol Futuring [11, 12], ProtoPoliSim trades the irreplaceable depth of human participants for repeatability, systematic variation, and scale—positioning it as a complement to, not replacement for, participatory methods. Yang et al.’s vision of HCI-policy collaboration [25] and Wong et al.’s infrastructural speculations [24] provide the broader framing: protocols are designable artifacts that deserve the same iterative prototyping and testing that HCI applies to interfaces.

Limitations. Several methodological challenges must be acknowledged. First, the system’s “detection” of second-order dynamics is more accurately described as *structured elicitation*: LLM agents annotate their own outputs with dynamic types, which is generation rather than independent verification. Unlike the historical infrastructure failures documented by Star and Ruhleder [21] or the empirical ossification patterns Ammar [1] observed in Internet protocols, our elicited dynamics have no ground-truth validation against real deployments. Second, LLMs exhibit tendencies toward sycophancy and mode collapse [2, 17] that may limit genuine adversarial pressure; even generative agents with rich memory architectures can converge on shared behavioral patterns, risking predictable rather than genuinely creative exploits. Third, persona diversity is bounded by training data. While Paglieri et al.’s Persona Generators [16] optimize for support coverage and Wang et al.’s DeepPersona [23] scales attribute-rich profiles, both inherit the biases of their underlying LLMs—a concern especially acute for protocols affecting populations underrepresented in English-language corpora. Fourth, Clark et al.’s “tussle” framing [6] presupposes that protocol conflicts are legible as discrete stakeholder positions, which may miss diffuse forms of resistance that infrastructure ethnographers have documented [20].

Open questions for the PoliSim community. We see three directions for engagement. First, *benchmarking*: can the community develop shared protocol scenarios—with known historical second-order dynamics—against which simulation outputs can be validated? Second, *hybrid workflows*: how should automated outputs feed into participatory governance, serving as reconnaissance rather than replacement for stakeholder engagement? Third, *persona calibration*: what methods can ensure synthetic populations cover affected stakeholders’ actual diversity?

5 Conclusion

We have argued that governance protocols need pre-deployment stress-testing that can surface second-order dynamics, and that LLM agents grounded in deep synthetic personas and structured by the Protocol Futuring adversarial cycle offer a viable mechanism. ProtoPoliSim is our working prototype—not a finished tool but an invitation to the PoliSim community to test it on your own governance cases and to help develop the validation frameworks this approach demands.

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